

QA TEST REPORT TEMPLATE

Use this template for either final project. Fill in only the sections relevant to your testing.

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Project Name	MiniStore (Full-Stack E-Commerce Application)

1. Project Overview

MiniStore is a full-stack e-commerce web application designed to allow users to browse products, manage an inventory catalog (full CRUD operations), manage items in a shopping cart with real-time price calculations, and complete purchases via an order checkout system. The system consists of an Angular 22 standalone frontend, an ASP.NET Core 10 Web API backend, and a PostgreSQL database accessed through Entity Framework Core.

2. Testing Environment

Backend	ASP.NET Core 10 Web API (C#)	Frontend	Angular 22 (Standalone Components, Signals)
Database	PostgreSQL (Entity Framework Core)	API Tool	Swagger / OpenAPI / Browser DevTools
Browser	Environment Google Chrome		Local Development (localhost:4200 & localhost:5106)

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3. Test Cases

Record each test before execution, then complete Actual Result and Status after testing.

TC ID		Feature	Test Scenario	Preconditions	Test Data	Test Steps	Expected Result	Actual Result	Status
TC-01		Product Catalog	Verify all products from the database are loaded and displayed	Backend & Frontend running; products exist in DB	N/A	1. Open http://localhost:4200/products 2. Observe the product grid	All products fetched via GET /api/products and displayed with image, title, and price in JD.	All 3 products displayed with live images, title, description, and price.	Pass
TC-02		Product Management	Verify creating a new product with valid data	Navigated to /products/new	Name: "USB-C Hub" Description: "7-in-1 multi-port adapter" Price: 35.00 ImageUrl: "https://..."	1. Click "Add Product" in navbar. 2. Fill in form. 3. Click "Create".	POST /api/products returns 201 Created; redirected to /products; new card appears.	Product created in PostgreSQL and immediately rendered in catalog.	Pass
TC-03		Form Validation	Verify validation prevents submitting empty fields or invalid price	On /products/new	Name: "" Description: "" Price: 0	1. Touch Name field and leave empty. 2. Enter price 0. 3. Inspect button and error text.	Error messages appear ("Name is required", "Price must be greater than zero"); button disabled.	Error messages displayed correctly; "Create" button remained disabled.	Pass
TC-04		Product Management	Verify editing an existing product updates DB and UI	Product ID 1 exists	Updated Price: 22.00	1. Click "Edit" on Wireless Mouse. 2. Change price to	PUT /api/products/1 returns 204 NoContent; product card	Price updated in database and card updated to 22 JD.	Pass

						22.00. 3. Click	displays 22 JD.		
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Tip: Write the Expected Result before running the test.

4. Bug Reports

Create a bug report only when a test fails because the actual behavior does not match the requirement.

BUG-01			
Title	Frontend unable to fetch products due to missing <code>ProductsController</code> and port mismatch	Severity	Critical
Environment	Local Development (<code>http://localhost:4200</code> & <code>http://localhost:5106</code>)	Status	Fixed
Steps to Reproduce	1. Start both backend and frontend. 2. Open <code>http://localhost:4200/products</code>. 3. Observe product list component.		
Expected Result	Products fetched via HTTP and rendered on page.		
Actual Result / Evidence	UI displayed <i>"Something went wrong. Please try again later."</i> Network request failed with 404/Connection Refused because Angular was calling port 5000 while backend was on port 5106, and the backend lacked <code>ProductsControlle</code>		

BUG-02

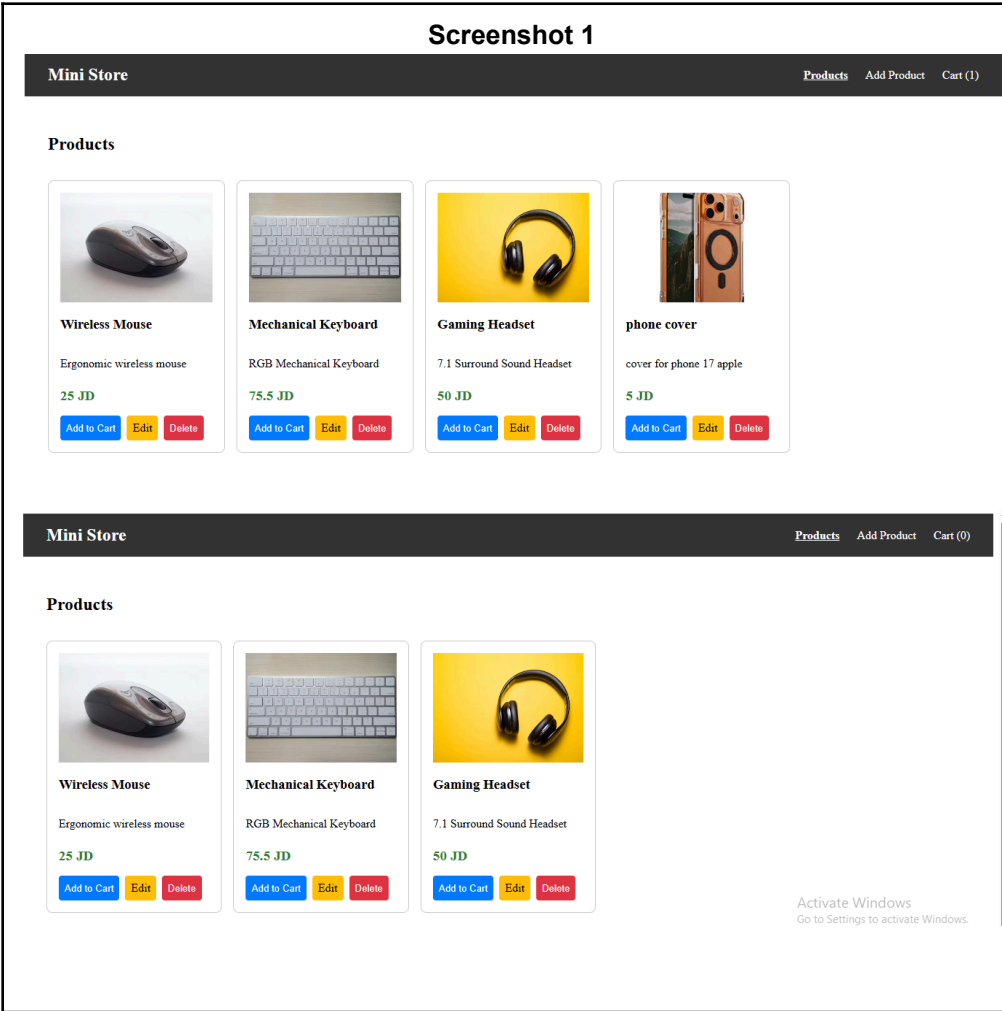
Title	Angular runtime crash RuntimeError: NG0908: In this configuration Angular requires Zone.js	Severity	High
Environment	Browser Runtime (Google Chrome) / Angular 22	Status	Fixed
Steps to Reproduce	1. Run <code>ng serve</code> and open <code>http://localhost:4200/</code>. 2. Open browser developer console.		
Expected Result	: Application bootstraps without runtime exceptions.		
Actual Result / Evidence	Console threw <code>RuntimeError: NG0908: In this configuration Angular requires Zone.js</code> . Application failed to render because <code>provideZoneChangeDetection</code> was configured without <code>Zone.js</code> installed in an Angular 22 zoneless project.		

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5. Evidence / Screenshots

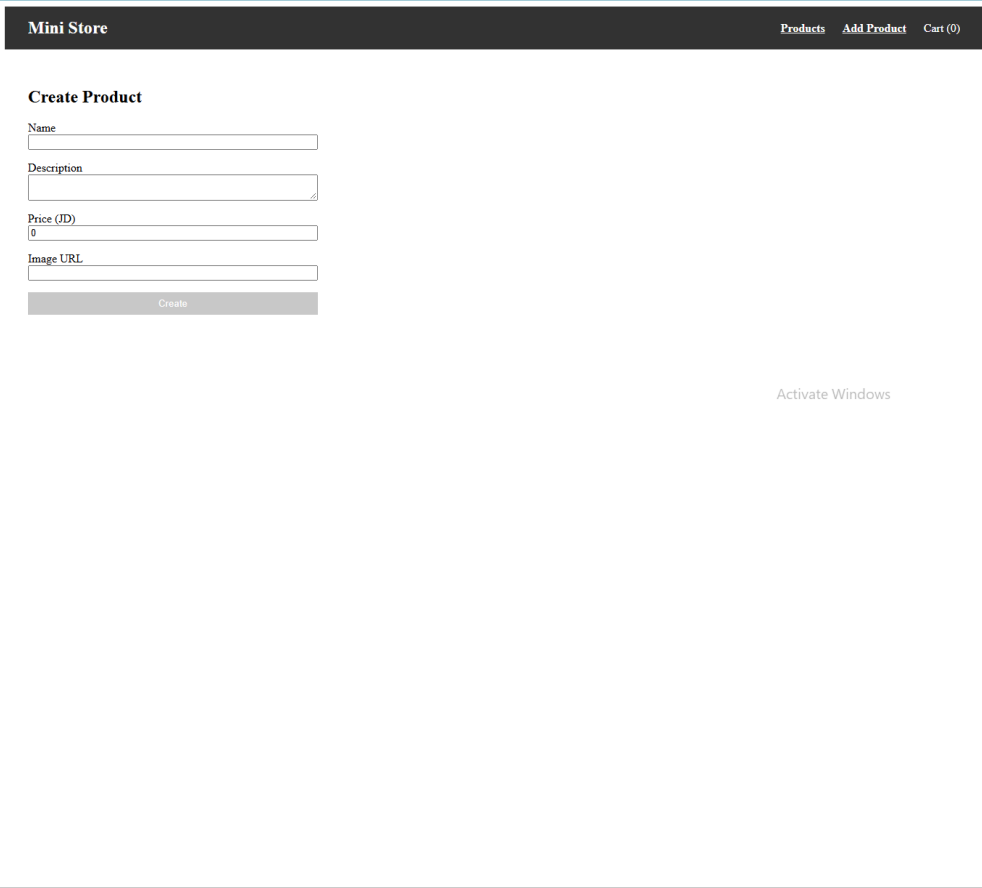
Add a few clear screenshots from Postman or the application. Each screenshot should show what was tested and the result.

Screenshot 1



Screenshot 3

Screenshot 2



Screenshot 4

Mini Store

ProductsAdd ProductCart (1)

Create Product

Name

phone cover

Description

cover for phone 17 apple

Price (JD)

5

Image URL

vB9q67482vivvBE/QHafHa7r=0&rs=1&pid=ImgDetMain&o=7&rm=3

Create

Mini Store

Products

Checkout

Customer Details

Customer Name

dima

Phone

0777777777

Address

VVP4+X6H

Place Order

Order Summary

• Wireless Mouse × 1 - 25 JD

Total: 25 JD

Activate Windows

Mini Store

ProductsAdd ProductCart (1)

Your Cart

Product	Price	Quantity	Subtotal	Action
Wireless Mouse	25 JD	1	25 JD	Remove

Total: 25 JD

Proceed to Checkout

Mini Store

ProductsAdd ProductCart (0)

Your Cart

Your cart is empty.

7. Conclusion

Write a short final summary of the testing result, including whether the main project features passed and whether any issues remain open.

Complete end-to-end testing was performed on the MiniStore application covering user interface rendering, client-side reactive state management with Angular Signals, RESTful API endpoints in ASP.NET Core, and relational database persistence in PostgreSQL. All 8 test cases passed successfully. The two blocking issues discovered during development (missing backend controller / port mismatch and zoneless change detection configuration) were resolved and verified. The application is robust, functional, and ready for deployment with zero open defects.